Kubernetes Architecture in EKS (Hinglish) 🔥

In AWS EKS, AWS manages the **Control Plane** for you.

You mainly manage:

- worker nodes
- pods
- deployments
- services

AWS handles the difficult master/control-plane components.

Big Picture First 🧠

Kubernetes has 2 major parts:

Part	Purpose
Control Plane	Brain
Worker Nodes	Actual work

In AWS EKS

AWS Manages:

✅ API Server
✅ etcd
✅ Scheduler
✅ Controller Manager
✅ Control Plane HA
✅ Patching
✅ Availability

You Manage:

✅ EC2 worker nodes
✅ Pods
✅ Deployments

✅ Networking configs
✅ IAM/RBAC
✅ Apps

Control Plane Components Explained 🔥

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1. kube-apiserver

MOST IMPORTANT component.

Everything talks to API Server.

Think of it as:

> “Main entry gate of Kubernetes.”

Example

When you run:

```
kubectl get pods
```

Actually:

- kubectl → API Server
- API Server → etcd
- Response comes back

In EKS

AWS hosts API server for you.

You don’t install/manage it.

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2. etcd

etcd = Kubernetes database.

Stores cluster state.

Stores Things Like:

- Pods
- Deployments
- Secrets
- ConfigMaps
- Node info
- Cluster configs

Example

If pod created:

```
nginx-pod
```

Its info stored inside etcd.

Important ⚡

If etcd dies completely:

- cluster brain lost 💀

That’s why etcd is critical.

In EKS

AWS manages:

- backups
- replication

- HA
- recovery

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3. kube-scheduler

Scheduler decides:

> "Which node should run this pod?"

Example

Suppose:

- 4 nodes available
- pod created

Scheduler checks:

- CPU
- memory
- taints/tolerations
- affinity
- nodeSelector

Then chooses best node.

In EKS

AWS manages scheduler.

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4. kube-controller-manager

This watches cluster continuously.

It tries to maintain desired state.

Example

Deployment says:

```
replicas = 3
```

But one pod crashes.

Controller manager notices:

```
Current = 2
Desired = 3
```

It creates another pod automatically 🔥

Types of Controllers

- Node controller
- Replication controller
- Endpoint controller
- Job controller

etc.

In EKS

AWS manages it.

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5. cloud-controller-manager

Special controller for cloud providers.

Connects Kubernetes with AWS services.

Example

When you create:

```
type: LoadBalancer
```

Cloud controller automatically creates:

- ELB/ALB/NLB
- AWS networking configs

Without Cloud Controller

Kubernetes can't talk to AWS APIs.

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6. kubelet

Runs on EVERY worker node.

Very important.

Think:

> "Node agent."

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kubelet Job

It talks with API server and says:

```
"I'll run these pods."
```

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kubelet Responsibilities

✅ Pull images
✅ Start containers

✅ Monitor pods
✅ Report node health

Example Flow

1. Scheduler selects node
2. kubelet sees assigned pod
3. kubelet asks container runtime to start it

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7. kube-proxy

Handles networking inside cluster.

Runs on every node.

Main Job

Provides Service networking.

Example

You create Service:

```
ClusterIP
```

kube-proxy creates iptables/ipvs rules.

Then traffic reaches correct pods.

Example Flow

User → Service IP → kube-proxy → Pod

Without kube-proxy

Services won't work properly.

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8. kubectl

NOT control plane component.

It's just CLI tool.

Purpose

Used by humans/admins.

Example:

```
kubectl get pods
kubectl apply -f app.yml
kubectl describe pod nginx
```

kubectl communicates with API server.

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9. kubeadm

Very important interview topic ⚡

kubeadm is used to CREATE Kubernetes cluster manually.

Example

In self-managed Kubernetes:

```
kubeadm init
```

creates:

- API server
- scheduler
- etcd
- certificates
- control plane setup

Worker Join

```
kubeadm join
```

adds worker node.

BUT IN EKS ⚡

AWS already creates control plane.

So usually:

- you DO NOT use kubeadm in EKS

AWS handles cluster bootstrap.

Simple Difference

Component	Purpose
kubectl	CLI tool
kubeadm	Cluster creation tool
kubelet	Node agent
kube-proxy	Networking
etcd	Database
scheduler	Pod placement
controller-manager	Desired state management

Component	Purpose
API server	Main gateway

Easy Memory Trick 🧠

Component	Remember As
API Server	Receptionist
etcd	Database
Scheduler	Decision maker
Controller Manager	Auto-healing manager
kubelet	Node worker
kube-proxy	Traffic manager
kubectl	User remote
kubeadm	Cluster installer

EKS Architecture Reality 🔥

AWS Handles

✅ Control Plane
✅ etcd
✅ HA
✅ Upgrades
✅ API server

You Handle

✅ Nodes
✅ Pods
✅ Applications
✅ YAMLs
✅ Scaling policies
✅ Security configs

Interview One-Liner 🎯

> “In EKS, AWS manages the Kubernetes control plane components like API server, etcd, scheduler, and controller manager, while users mainly manage worker nodes and workloads.”